CVCITC
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Barangay E-complaint and Management System

A Capstone Project

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CHAPTER I
THE PROBLEM AND ITS BACKGROUND

I. Introduction

Nowadays, where the computer and the technology is very critical and online communication is part of living, the people who are using it all over the world are different. Most of the barangays here in Santiago City are using paper-based material in keeping all the records of a resident, but they all know that day by day the population of their barangay is increasing. In the past, it was easy to use paper because there were fewer residents and it was easier to manage them. But as the years went by, the population got bigger just like in one of the barangay in Santiago City- Centro East which they couldn't handle all the request and complaint of the residents and it this made it difficult to manage because of bigger population and many complaints. Centro East, formerly Poblacion, is one of the barangays in the City of Santiago, Province of Isabela. It has 905 population as determined by the 2020 Census.

Many people would seek several information they require that can be found in the barangay. With only manual processing it would take time to search for keywords to retrieve content. But with the help of computers it would greatly enhance the time and end result you will get in comparison to manual time or work done by hand. According to Lorenzo (2015), one of the problems of

barangay officials for providing timely response to constituents is the lack of facilities, materials and resources, consequently, the performance of the barangay officials is adversely affected by this inadequacy of non-human resources. The use of modern computer technologies will accelerate the reform process by assuring the standardization of the procedures on the level of the entire system, their unification, as well as by including some indicators for measuring the efficiency of the legal actions (Stefan, I. 2006), hence, to truly fulfill its function of delivering justice and aid to its people, Barangay E-Complaint needs to have adequate facilities that will help in the ease of information access and efficient processes for the people.

Computerization is a control system that manages processes in the industrial workplace. It reduces human errors and processing time; thus, it can boost productivity and result in a high quality of production. This can result in a system well integrated process that can perform much faster and more accurate than the manual system (Martinez, et. al, n.d). In order to correct disputes, reconcile people and address concerns of the local community more efficiently, the Barangay E-Complaint System needs to adopt to the advancement of technology. Information System involves a variety of information technologies such as computers, softwares, communication systems, the internet, mobile devices and much more to perform specific task interact with and inform various

actors in different organizational or social context of general context to the field of Information system are all aspects of the development, deployment implementation, use and impact of it in organizations and society (Boell & Kecmanovic, 2015).

With this, Barangay E-Complaint Management System is propose. An E-Complaint Management System that will help residents in barangay Centro East to file a complaint through online transaction. In this study, the researchers want to allow the barangay Centro East the ability to avoid paperwork transaction, for the furtherance of barangay justice system and improve the local community so that they can serve their residents more easily.

II. Review of Related Literature and Related Studies

INFORMATION SYSTEM

Many Governments of the world have started e-Government strategies to renew the public sector in order to promote the rapid step of technological change. In line with e-Government plans and with the help of ICT, Government Agencies (Gas) are increasingly called upon to reduce costs and to deliver better public services tailored to the needs of the citizen and business. Sharing information and functions of the different legacy Information Systems (IS), in local and central GA's, is fundamental in a plan focusing on the improvement of the quality of services through new technologies. Government Information Systems are a complex environment that must be reused rather than replaced because replacement is cost-prohibitive. To achieve this goal, government have started projects foro the development of interoperability frameworks (Castilliano, Pastore & Arcieri, 2005) Inexpensive and powerful microcomputers, telecommunications technologies, and information systems have become an essential element of the development process of developing countries. Information systems for governance and public administration, as one of the most important aspects of computer applications in developing countries, have been a high priority concern in almost all developing countries. Governance and public administration cannot be productive, effective and efficient without the support of modern information

technology. The use of information technology has been an absolute requirement for public administration and management development. The governance and Public Administration and Development Management of UN/DDSMS has been continuously and increasingly under pressure from developing countries to provide advisory services and technical assistance in this important area (United Nations, n.d)

Through e-government, an increasing number of nations are making major commitments to modernize and reform government in an effort to achieve greater efficiency, broaden access to government services and improve service levels. Technology is seen as a key enabler for e-government (Elmagarmid & McIver, 2001). Ultimately, e-government seeks to centralize and make a cohesive and seamless set of government services available to end-users. Hence, the integration of information systems (IS), both within and between different government departments, will become a critical issue as ee-government matures (Golden, Hughes, & Scott, 2003). As e-government matures, information systems (IS) integration will become critical to the ability to provide centralized and seamless online services (Lam,2008)

Acording to Schware & Deane (2003) in the study “Deploying e-government programs: the strategic importance of “I” before “E”, many developing countries are in the initial phases of adopting electronic government

(e-government) programs to improve public services and deliver them as efficiently and conveniently as possible. Their experience with a variety of governments throughout the developing world at different stages of implementing e-government programs with citizens (G2C), businesses (G2B), and other entities of government (G2G) suggests that a major reason behind the success or failure of e-government projects is the extent to which, first, the governments address technological infrastructure encouraged by appropriate telecommunciations policies; and second, the legal and regulatory instruments required for e-government. Information and communication technology (ICT) infrastructure (the "I") development is at the heart of successful deployment and sustainability of e-government programs.

According to Hussein, et. Al (2007) in the study entitled "The Influence of Organizational Factors on Information Systems Success in E-government Agencies in Malaysia", Information system success continues to be a subject of interest among IS researchers. The literature nevertheless offers limited understanding in regard to organizational factors influencing IS success particularly in public sector organizations. Using perceptual measures, this study aims to investigate the influence of organizational factors on IS success in selected public sector agencies in Malaysia. Data was gathered from 201 users from four central agencies located at the central administration complex in

Putrajaya. Based on related studies in the literature, six organizational factors and four IS success dimensions were used to investigate the relationship between the identified organizational factors and IS success dimensions within the scope of the selected e-government agencies. The organizational factors are top management support, decision-making structure, management style, managerial IT knowledge, goal alignment, and resources allocation, whereas the IS success dimensions include systems quality, information quality, perceived usefulness, and user satisfaction. The study found that all the organizational factors are significantly correlated with the four IS success dimensions under investigation. Further analyses reveal that goal alignment is the highest predictor of IS success, followed by management style and centralization of decision-making structure. Least expected was the top management support factor which appear to be the weakest predictor of IS success is insignificant. The study concludes that the findings are generally consistent with previous studies. On a specific note related to goal alignment, the public sector agencies can be seen experiencing a similar iimpact to that of their private counterparts in terms of IS strategy alignment, though they still generally subscribe to the traditional and centralized decision-making structure.

III. Statement of the Problem

This study seeks to develop a complaint system for the barangay of Centro East, Santiago City which intends to help the barangay officers in charge in the particular office in the ease of access of information and processes for a more efficient render of service in the community of jurisdiction improving the barangay justice service.

This study aims to develop a Barangay E-complaint Management system. Specifically, it seeks to present the following:

1. What are the practices and problems encountered in the existing Barangay complaint system?
2. What system can be developed to address the problems in the Barangay complaint system?
3. What is the extent of compliance of the developed system to ISO 25010 Software Quality Satndards as assessed by IT experts in terms of?

   3.1 Functional Sustainability

   3.2 Performance Efficiency

   3.2 Compatability

   3.3 Usability

   3.4 Reliability

3.5 Security

3.6 Maintainability; and

3.7 Portability

4. What enhancement can be made to improve the developed system?

IV. Research Paradigm

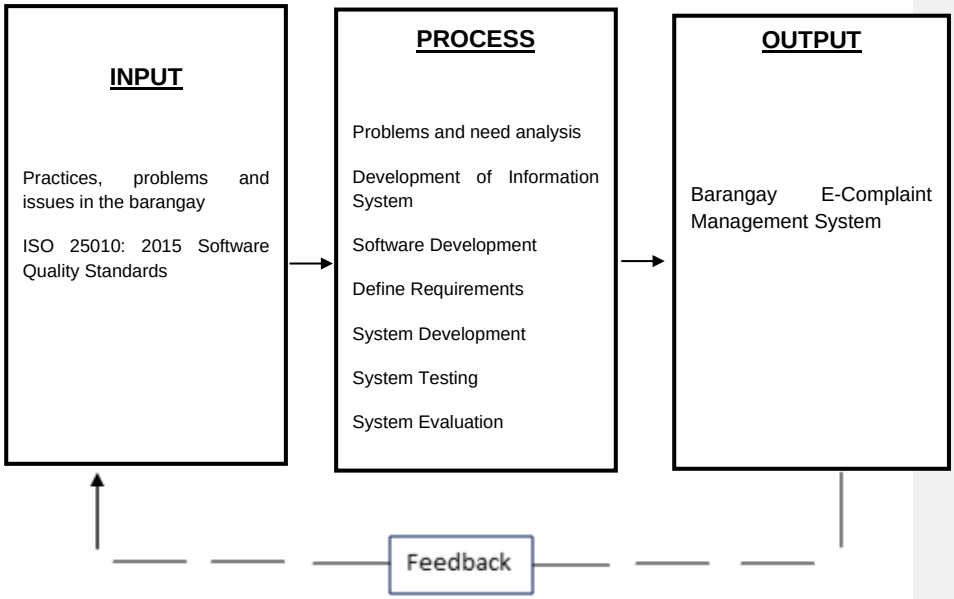

Figure 1.Research Paradigm

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INPUT
Practices and Problems Encounterd

Forms Complaint

Process and procedures in Filing Complain in the Barangay

OUTPUT

Resuls of Evaluations (include)

The figure above presents the research paradigm of the study. The study will revolve under the Input Process Output framework in developing an information system intended for the barangay e-complaint management system. The input of the study will include the practices and problems encountered by a citizen in a certain barangay. Additionally, the input will integrate the ISO standards of 2015 for software quality standards. The process of this study will involve an analysis of the identified problems and practices for the development of the system, practical test, deliver and system evaluation. After the aforementioned procedures, the expected output will be the barangay E-Complaint Management System.

V. Significance of the Study

The development of Barangay E-Complaint System in Barangay Centro East will be of great help to the citizens in the local premises of the Barangay as they will experience faster transaction and efficient services from the office for the automation of processes, more specifically the following:

The Barangay & its officials. This system will bring long-term benefit to the barangay as the system can be used by current and succeeding administrations and officials. The system will be of great help to the officers in terms of time

maximization and office procedures, it will enable easier processes of documents, filing, retrieval and issuance.

Barangay Residents (Complainant/Defendant). The system will be beneficial to them in terms of faster access to the needed services. Through this system, transactions will be accomplished at a shorter time, hence the clients will be catered at minimal time and maximum utility.

The Local Government Unit. Once proven effective, this will be a good model system for other barangays to replicate and implement, improving the services of all barangays within the city.

Researchers. The researchers will be benefit from the study; it makes them more knowledgeable in performing and increases their potential in conducting research for their thesis. This study will also be an additional resource or an essential reference in making the same study or in developing the same study.

Future Researchers. This study can be a basis for future extension studies or studies of similar domain.

VI. Project Highlights

Barangay E-complaint Management System is a system that will be effectively cater the complaints in the barangay. It will help the barangay officers in providing an efficient and effective system to their residents.

Digitization- the complainants may upload an image in the complaint system for the supporting details in their complain.

SMS- the system has an sms feature for the updates to the complainant about the status of his complaint.

VII. Scope and Limitation

This study aims to create a computerized system which focused on developing an E-Complaint System for the barangay complaint desk of barangay Centro East, Santiago City. This involves the recording procedures, practices and problems encountered by the officer-in-charge of the complaint desk and figure out a system that will be able to address the identified problems. More so, assessment on functionality, reliability, usability, efficiency, maintainability, portability wil be made to determine its marketability and acceptability as well. Owing to the aforementioned constraints, findings are applicable only to the areas that were covered by the study.

VIII. Definition of Terms

Barangay- A small territorial and administrative district forming the most local level of government.

Complaint- Refers to a statement expressing that something is unsatisfactory or unacceptable.

Complaint Desk- In the context of this study, this refers to the official desk of barangay justice system where residents seek for local mediation assistance over circumstances that requires the help of Lupong Tagapamayapa.

Data- A distinct pieces of information, usually formatted in a special way.

Digitization- The process of inserting image in the system.

Database- Collection of information organized in such a way that a computer program can quickly select desired pieces of data. A stored data of complainant and defendant informations, user, puroks, and officials

System- A set of connected things or devices that operate together.

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CHAPTER II
METHODOLOGY

I. Research Design

This study will use the developmental research design since it proposed a Barangay E-complaint Management System. The Barangay E-complaint management System is firmly analyzed to provide a good quality of service for the Barangay and its citizens. The system was created using the method and procedures of Rapid Application Development.

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Since you use the ISO, include descriptive research and SDLC in the Development. SDLC is the RAD

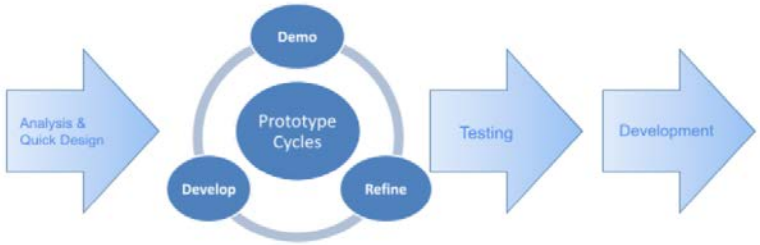

Figure 2. Rapid Application Development (RAD)

Analysis and Quick Design. The researchers will gather the data neded to fulfill the requirements for the system; The researchers gathered the data needed to

fulfill the requirements for the system; they decided on the quick design of the system.

Prototype Cycle. The Prototype Cycle has three components, namely: 1) the **Develop** Phase where the system was developed, 2) the **Demo** Phase, where the researchers demonstrated the developed system, and 3) the **Refine Phase** where the researchers defined and decided for the refinements and improvements of the system for better performance.

Testing. The developed system is now ready to be tested in the Barangay Centro East with its Residents.

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II. System Architecture

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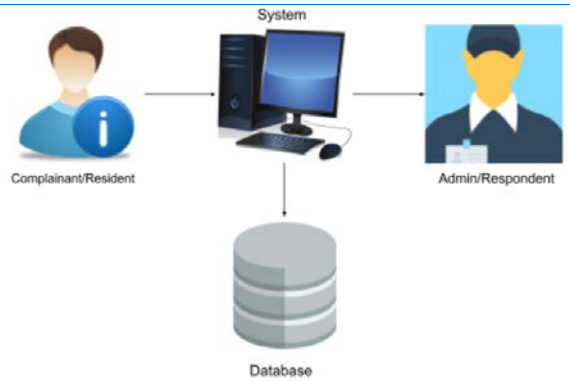

Figure 3. System Architecture

This figure illustrates the System architecture of the system. Complainant/Barangay Residents will use the system and input their credentials to verify their identity then followed by their complaint and after submission the admin/respondent will have a notification about the complainant's complaint at the same time the transaction and information will be stored in the database and upon completion of processing and analyzation of the complaint. The complainant will receive a notification based on the completed analysis and response to the complaint.

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III. Hierarchical Input Process Output

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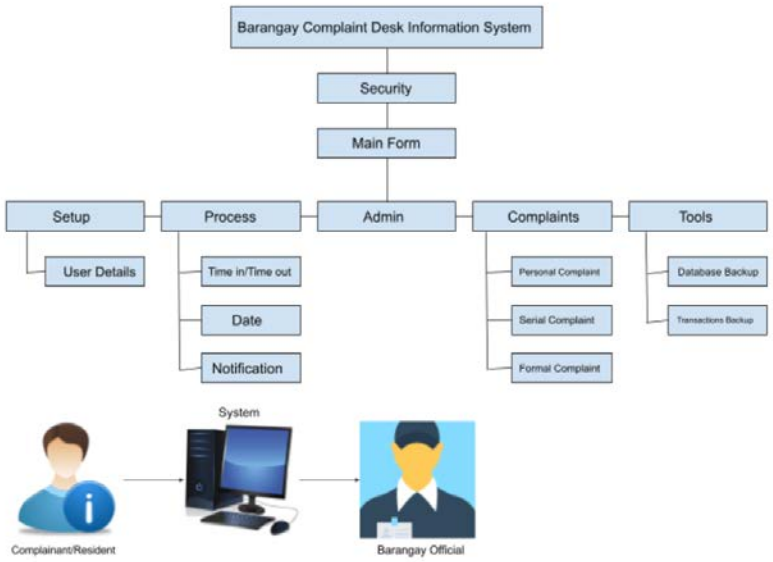

Figure 4. Hierchical Input Process Output Diagram

This Figure illustrates the Hierarchical Input Process Output of the system. It shows how the system works and the components of each of the processes as well as the overall design of the system being implemented and its specific requirements. The Setup module contains one sub module, the user details while the Process module contains three sub modules: the Time in/Time out , Date and Notification to the Complainant and Barangay Official. The Complaints module contains another three sub modules, the Personal Complaint, Serial Complaint and Formal Complaint. Finally, the Tools module contains two sub modules, the Database Backup and Transaction Backup.

IV. Software Requirements

Software	Requirement
Operating System	**Microsoft Windows 10**
Apache Server	**XAMPP**
Database Server	**MySQL**
PHP Maker	**2022 Version**

Table 1. Software Requirements

Commented [U7]: What is Programming Language and the version

Table 1 above shows what the software to be used, this software is commonly used and they're available online for installation.

V. Hardware Requirements

Hardware	Requirement
DESKTOP	
DESKTOP	
Motherboard	**GIGABYTE GA-H110-DS2 1151, DDR3, HDMI**
Memory RAM	**4GB DDR3 PC10600/1333 or Higher**
Hard Disk Drive (HDD)	**SP550 120GB SSD**
Processor	**Intel core i3 or higher recommended**

Table 2. Hardware Requirements

Table 2 shows the hardware requirements that is suitable for the proposed system since it is easy to use and has low-cost maintenance.

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VI. Locale of the Study

This study will be carried out in Santiago City in the municipality of Barangay Centro East where the resident of their community are commonly reside and work. Furthermore, to make the study more feasible, it will be conducted and

applied to their Barangay Officials as well to provide insights for the residents to make the system easier to use. At the same time, the written, vocal, and online complaints data gathering will be complied to the Barangay Officials to understand the feelings and behavior of their residents.

VII. Respondents of the Study

Respondents	No. of Respondents	Description
1. IT Practitioner	3	IT practitioner will determine the acceptability of the system.
Total Respondents	3 individuals	

Table 3. Respondents

The table above represents the respondents, the developers used an interview guide and purposive sampling technique to produce a sample representative most often from a population that is difficult to measure. Three IT experts whom will test the development of the system, with a total of (3) three respondents.

VIII. Instrumentation

The instruments used in the study in gathering required information were interview with the participants of this study, document review, and questionnaire. The interview and questionnaire were used to guide the researchers in designing and developing the system.

The ISO 25010:2015 Software Quality Standards was used as questionnaire to evaluate the quality of the system being developed on its functionality, reliability, usability, efficiency, maintainability and portability of the system among end users.

IX. Data Gathering Procedure

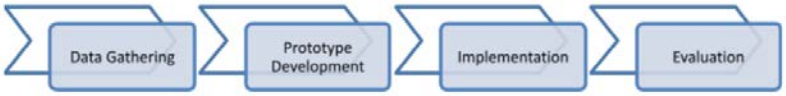

Figure 5. Prototype-Feedback-Improved Prototype

Data Gathering. Data needed in designing and crafting of the system was gathered. The gathering of data was done through meetings with the Administration, Students and Parents. Moreover, the internet was used for conceptualization of the system and overview.

Prototype Development. The prototype system that is intended for presenting the design and how the system works was crafted and presented to the administration of the school to identify and collect additional requirements needed for the improvement of the system. In addition, the construction of the system includes the creation of databases, application programs, and user and system interface.

Present System Workflow. First is the **Data Entry** which a user or complainant must input his credentials/identification/information and his/her complaint. Second step is the **Data Process** where the complaint will be analyzed and verify the complainant's identity. The last step is the **Output** where the complaint details are displayed on the screen and the information and date of inquiry is stored in the database. The system will then notify the barangay official or the person in charge of the system and notify the respondents for its response in the complaint.

X. Data Analysis

The weighted mean is used as the statistical tool. The weighted mean is computed to measure the system's compliance with ISO 25010:2015. The formula is written as;

Weighted mean = $\Sigma wx/\Sigma w$

Σ = the sum of.

w = the weights.

x = the value.

The system will be evaluated using the ISO 25010 Software Quality Characteristics. Thus, this type of research will only be using the survey form: use Weighted Mean and Likert Scale of 1-5: (5-Complant to the very high extent, 4-Compliant to a high extent, 3-Compliant to a moderate extent, 2-Compliant to a low extent, 1-Compliant to a very low extent)

Table 4. Likert Scale with Numerical Interpretation

Likert Scale	Weighted Mean	Description
5	4.01-5.00	Compliant to a very high extent
4	3.16-4.00	Compliant to a high extent
3	2.51-3.15	Compliant to a moderate extent
2	1.76-2.50	Compliant to a low extent
1	1.00-1.75	Compliant to a very low extent

CHAPTER III
RESULTS AND DISCUSSION

This chapter presents the results of the data gathering and analysis in terms of problems and issues, features of the developed system to address the mentioned difficulties, the acceptance level of the developed system, and the improvements for continuous and future development. The developed modules of Barangay Complaint Managements System undergoes ISO 25010 Software Quality Standards had been assessed by the IT Experts.

1. **The practices and problems encountered in the existing Barangay Complaint System.**

Maintaining an accurate complaint log book can be a tedious affair. When you have a complaint, the most important thing is that your complaint is heard and your problem is addressed. If complaints get lost, they can't be resolved. Good complaint management software will allow you to set up automatic acknowledgements. The Barangay E-Complaint Management System is a simple system that greatly simplifies and assist in the processing of complaint records.

Compared to the traditional manual logbook that takes up lot of space and resources and prone to being misplaced or damage. The Barangay E-Complaint Management System if far more reliable and efficient in doing the task. Most especially in making changes to the records and the security of the said records.

2. **Developed Web Application design that solved the identified problem and issues.**

The system was developed using the latest web development tools: PhpMaker2022 and MySql for its database. Distinct modules of the developed system were as follows:

Figure 6. Module for Resident Records Management

Resident Records Management. This module is used to manage the profiles and records of the resident.

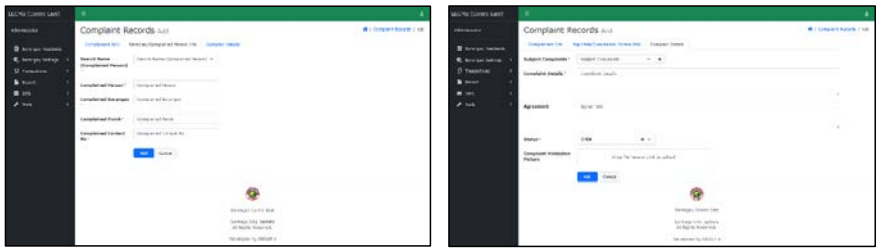

Figure 7. Module for Managing Complaints

Complaint Records. The barangay Secretary will be the one who will create the account and manage the complaints of the residents.

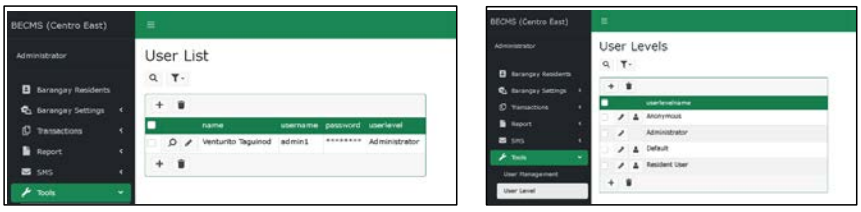

Figure 8. Module for Managing Users Access Level

System Users Management. The security module of the system is used to manage by the system administrator for registration and maintenance of users who will be the only ones allowed to access the system and create flexible user

level and its access permission to the system, the administrator can put restrictions.

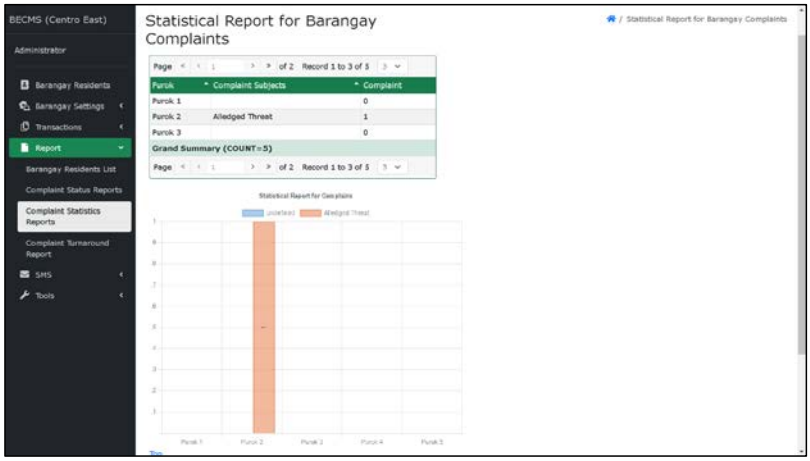

Figure 9. Barangay Complaint Statistical Report

Barangay Complaint Statistical Report. The system is capable of generating important statistical reports for top or most complaint in a Purok.

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3. **The extent of acceptance level of the developed system to ISO 25010 Software Quality Standards as assessed by the IT Experts.**

Table 5. IT Experts' acceptance level of the developed system in terms of Functional Sustainability

Indicators	Mean	Descriptive Equivalent
1. Functional Completeness	3	Compliant to a Moderate extent
2. Functional Correctness	3	Compliant to a Moderate extent
3. Functional Appropriateness	3	Compliant to a Moderate extent
Category Mean	**3**	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of functional sustainability. The system got a category mean of 3, descriptively interpreted as compliant to a moderate extent. The result implies that participants agreed that system functions met the specified tasks, requirements, and users' objectives provide accurate output, protection from unauthorized access and the processing time is reasonably fast.

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Table 6. IT Experts' Acceptance Level of the Developed System in Terms of Performance Efficiency

Indicators	Mean	Descriptive Equivalent
Time-Behavior	3	Compliant to a Moderate extent
Resource Utilization	3	Compliant to a Moderate extent
Capacity	3	Compliant to a Moderate extent
Category Mean	3	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Performance Efficiency. The system got a category mean of 3, descriptively interpreted as compliant to a moderate extent. The result implies that the system attained specification from users, produces desired output with optimum time, and processes required the least amount of hardware and perform greater useful work transactions.

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Table 7. IT Experts' Acceptance Level of the Developed System in Terms of Compatibility

Indicators	Mean	Descriptive Equivalent
Co-Existence	2.67	Compliant to a Moderate extent
Interoperability	2.67	Compliant to a Moderate extent

Category Mean	2.67	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Compatibility. The system got a category mean of 2.67, descriptively interpreted as Compliant to a Moderate extent. The result implies the system is compatible with different system configurations and specifications and compatible with different operating systems. This also shows that compatibility of the developed system as expected with the customer platform configuration is made.

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In your Repondents you select IT experts but in the results and discussion you use users

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Table 8. IT Experts' Acceptance Level of the Developed System in Terms of Usability

Indicators	Mean	Descriptive Equivalent
Co-Existence	2.67	Compliant to a Moderate extent
Interoperability	2.67	Compliant to a Moderate extent
Category Mean	2.67	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Usability. The system got a category mean of 2.67, descriptively interpreted as Compliant to a Moderate extent. The result implies that the system is user-

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centered, fit to use by different categories if end-user, easy to use and learn attractive and responsive graphical user interface.

Table 9. IT Experts' Acceptance Level of the Developed System in Terms of Reliability

Indicators	Mean	Descriptive Equivalent
Maturity	3	Compliant to a Moderate extent
Availability	3	Compliant to a Moderate extent
Fault Tolerance	2.67	Compliant to a Moderate extent
Recoverability	2.67	Compliant to a Moderate extent
Category Mean	2.84	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Reliability. The system got a category mean of 2.84, descriptively interpreted as Compliant to a Moderate extent. The result implies that the system failure-free operation for a specified period in a specified environment. The system performs its intended functions and operations in a system's environment, without experiencing a system crash.

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Table 10. IT Experts' Acceptance Level of the Developed System in Terms of Security

Indicators	Mean	Descriptive Equivalent
Confidentiality	2.33	Compliant to a Low extent
Integrity	2.67	Compliant to a Moderate extent
Non-Repudiation	2.67	Compliant to a Moderate extent
Accountability	2.67	Compliant to a Moderate extent
Authenticity	3	Compliant to a Moderate extent
Category Mean	2.67	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Reliability. The system got a category mean of 2.67, descriptively interpreted as Compliant to a Moderate extent. The result implies the system is protected with unauthorized access.

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In your Repondents you select IT experts but in the results and discussion you use users

check

Table 11. IT Experts' Acceptance Level of the Developed System in Terms of Maintainability

Indicators	Mean	Descriptive Equivalent
Modularity	3	Compliant to a Moderate extent

Reusability	2.67	Compliant to a Moderate extent
Analyzability	3	Compliant to a Moderate extent
Modifiability	3	Compliant to a Moderate extent
Testability	3	Compliant to a Moderate extent
Category Mean	2.93	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Maintainability. The system got a category mean of 2.93, descriptively interpreted as Compliant to a Moderate extent. The result implies the system can be easily customized, continues functioning if changes are made, the system can easily be tested and deficiencies can be easily diagnosed from the system.

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In your Repondents you select IT experts but in the results and discussion you use users

check

Table 12. IT Experts' Acceptance Level of the Developed System in Terms of Portability

Indicators	Mean	Descriptive Equivalent
Adaptability	3	Compliant to a Moderate extent
Installability	3	Compliant to a Moderate extent
Replaceability	3	Compliant to a Moderate extent
Category Mean	3	Compliant to a Moderate extent

This table shows the assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Portability. The system got a category mean of 3, descriptively interpreted as compliant to a Moderate extent. The result implies the system can be easily customized, continues functioning if changes are made, the system can easily be tested and deficiencies can be easily diagnosed from the system. The result implies the system is easy to install, transferable from one computer to another without much effort, can run on windows operating systems without explicit requirements, and be easily integrated into another environment with consistent functional correctness. Portability saves time for anyone involved in moving new versions of the software across environments.

Commented [U19]: Check your Respondents

In your Repondents you select IT experts but in the results and discussion you use users

check

Table 13. Summary of IT Experts' Acceptance Level of the Developed System in Compliance with ISO 25010 Software Quality Standards.

ISO 25010 Software Indicators	Mean	Descriptive Rating
Functional Sustainability	3	Compliant to a Moderate extent
Performance Efficiency	3	Compliant to a Moderate extent
Compatibility	2.67	Compliant to a Moderate extent
Usability	2.67	Compliant to a Moderate extent
Reliability	2.84	Compliant to a Moderate extent
Security	2.67	Compliant to a Moderate extent

Maintainability	2.93	Compliant to a Moderate extent
Portability	3	Compliant to a Moderate extent
GRAND MEAN	**2.85**	Compliant to a Moderate extent

This table shows the summary assessment of users on the acceptance level of the developed system using ISO 25010 Software Quality Standards in terms of Functionality Sustainability, Performance Efficiency, Compatibility, Usability, Reliability, System Security, System Maintainability and System Portability. The system got a grand mean of 2.85, descriptively interpreted as Compliant to a Moderate extent. The result implies that the system is highly compliant with all of the indicators of ISO 25010 Software Quality Standards thus accepted by the users with high satisfaction.

Commented [U20]: Check your Respondents

In your Repondents you select IT experts but in the results and discussion you use users

check

4. Enhancement can be made to improve the system.

With the basis of the findings of the study, the following are some of the recommendations offered to our project to further improve the performance of our system:

The researcher should include “referral report” in case problem cannot be solved within barangay. A summary report should also be included as reference. Also make a manual or Guideline, which discusses that the Barangay Secretary

has the full access to the system. And last add features for the recording of case investigation/ proceedings with uploading of attachment.

CHAPTER IV
SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusions, and recommendations based on the results of the study.

A. Summary of Findings

1. Barangay Complaint system in the barangay is administered using the traditional pen-paper approach. When you are going to complaint in the barangay you will state your complaints manually and these might take some time writing in the log-book. Also the barangay needs to spend some time arrange all the file and record of complaints manually. Barangay E-complaint Management System is a system that will be developed to overcome the above statement problem. This study was

developed to provide a computerized complaint system in the Barangay of Centro East.

2. To address the identified features, the Barangay E-Complaint Management System was developed. The developed system was build using the latest web development tools for easier accessibility and usability. Distinctive features of the system include digitization, recording of files and also profile management.
3. The system got a grand mean of 2.85, descriptively interpreted as Compliant to a moderate extent result from the evaluation of IT Experts using ISO 25010 Software Quality Standards. All the eight (8) categories: System Functionality, System Efficiency, System Usability, System Reliability, System Portability, System Maintainability, System Security, and System Compatibility got a Compliant to Moderate extent to Compliant to High Extent.

B. Conclusions

Based on the findings, the developed system was significant to the Barangay Centro East in terms of data recording. The system provided paperless transactions, faster and better data collection. Different features and functionality have met acceptable quality standards. Feedback from the user has proved the system is usable and addresses the needs of the

Barangay E-complaint Management System in terms of data recording or management records. The researcher should also focus on the improvement of the different criteria based on the ISO 25010 Software Quality Standards that got lowest mean for the betterment of the system.

C. Recommendation

The researcher recommends the following for the improvement of the currently developed systems:

1. The system is highly recommended for continuous development for improvement, in order to increase its effectiveness.
2. Include new features to further enhance its performance.
3. Implement the system in its designated area to test its functionality and;
4. Apply generic design in order to cater more barangays.

Commented [U21]: efficiency, effectiveness. Check the categories in the criteria where the scores were very low

Commented [U22]: What particular features? Be specific

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Commented [U23]: Follow APA format

Appendices

APPENDIX A. EVALUATION TOOL

QUESTIONNAIRE

This questionnaire presents the ACCEPTABILITY of the proposed Barangay E-Complaint Management System. Please tick the relevant boxes as honestly as you can, this is a highly confidential. The contents of this questionnaire are absolutely confidential and no other individual will able to see the data which you will be given. The answers to these questionnaires will proceed immediately to the project researcher (Rodalita C. Mando, Venturito B. Taguinod, Radley M. De Asis, and Jessa Mae Laguerta) and to our project adviser (Mr. Winston G. Domingo)

I. Personal Data

Name (Optional)._____

Please rate each of the items with the scale shown below:
5- Strongly Agree, 4-Agree, 3- Neither, 2- Disagree, 1- Strongly Disagree

Criteria	1	2	3	4	5
FUNCTIONALITY					
The system does what is appropriate (Adequacy)					
The system has the available function required for its execution (Adequacy)					
The system does what was proposed correctly (Adequacy)					
The system is precise in executing its function (Adequacy)					
The system is precise in its result (Adequacy)					
The system interacts with the specified modules (Interoperability)					
The system has capacity for multiuser processing (Interoperability)					
The system has capacity to operate with networks (Interoperability)					
The system has secure access through passwords (Secure Access)					
The system has internal backup routine (Secure Access)					
TOTAL					
RELIABILITY					
The system don't have frequent failures (Maturity)					

The system reacts appropriately when failures occur (Tolerance to failures)					
SSThe system informs the user concerning invalid data entry (Tolerance to failures)					
The system is capable of recovering data in the event of failure (Recoverability)					
TOTAL					
USABILITY					
It is easy to understand the concept and application (Intelligibility)					
It is easy to perform its functions (Intelligibility)					
It is easy learn how to use (Learnability)					
The systems facilitates the users' data entry (Learnability)					
The systems facilitates the users' retrieval of data (Learnability)					
It is easy to operate and control (Operability)					
The system provides help in clear manner (Operability)					
TOTAL					
EFFICIENCY					
The systems response time is appropriate (Time)					
The system's execution time is appropriate (Time)					

The resources used is appropriate (Resources)					
TOTAL					
MAINTAINABILITY					
It is easy to find a failure when it occur (Analyzability)					
It is easy to modify and adapt (Modifiability)					
There is no great risk when changes are made (Stability)					
Changes are easy to test (Testability)					
TOTAL					
PORTABIILITY					
It is easy to adapt to other environment (Adaptability)					
It is easy to install in other environment (Capacity to be installed)					
It is easy to use to replace another program (Capacity to be replace)					
TOTAL					

Comments:

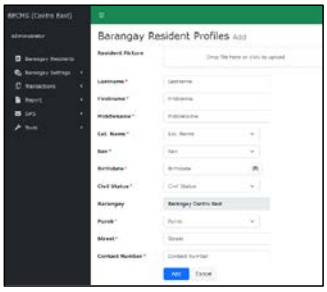
Recommendation:

_____
Respondent's Signature over Printed Name

APPENDIX B. User Guide/Manual

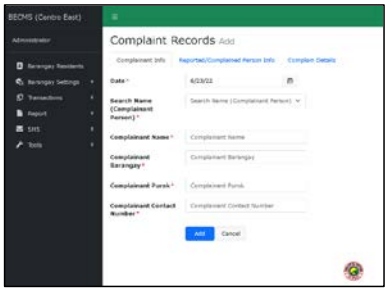
Step 1:

Log-in to the system using the username and password.



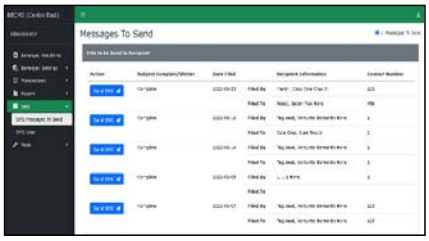
Step 2:

Fill-out the form for the Profile of the Residents.



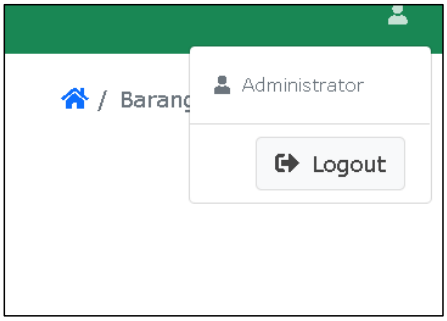
Step 3:

Right after you made the profile of the resident, you can fill-out the complaint record form.



Step 4:

After the process of filing a complaint, the Secretary will send a message for the follow-up of the complaint.



Step 5:

Log-out into the system.

CVCITC

Cagayan Valley Computer and Information Technology College

#28 Carreon St., Centro East, Santiago City, Philippines

MANDO, RODALITA C.

Address:
Plaridel, City of Santiago, Isabela
Birthdate:
September 15, 1999
Phone:
0909-664-0525
Email:
casirayarodalita@gmail.com

Skill Highlights

- Possesses Leadership Skills
- Computer Literate
- Ability to work independently or as part of the group.
- Can work under pressure
- Willing to be trained for improvement

Language

English
Filipino
Ybanag

Work Experiences

- Work Immersion
  Human Resource Management Office at Local Government Unit of City of Santiago
- On-The-Job Training
  College of Information Technology and Engineering office at CVCITC Centro East, City of Santiago.

Trainings and Webinars

- "Web Development using NuxtJS AND Vue.js"
  College of Computing Studies, Information and Communication Technology (CCSICT) Isabela State University - Echague, Isabela - April 02, 2022
- "Flourishing the True Spirit of Genuine Leaders"
  Lower Magat Eco-Tourism Park, Diadi Nueva Vizcaya
  October 4-5, 2019
- "Breaking Boundaries Amidst Diversity: Reaching the CVCITCians Goal"
  Mt. Atio-an Ramon Isabela
  September 25-26, 2018
- ROADSHOW @ 9th: ROAD to Leading an Vibrant Country, Show it through the Engineering Technology
  Isabela State University, Echague Isabela
  March 15, 2018

Education

Primary: Dubinan Elementary School
Dubinan, City of Santiago
2006 - 2012

Secondary: Cagayan Valley Computer and Information Technology College Inc.
Centro East, City of Santiago
2016 - 2018

Tertiary: Cagayan Valley Computer and Information Technology College
Centro East, City of Santiago
2018-2022

CVCITC

Cagayan Valley Computer and Information Technology College

#28 Carreon St., Centro East, Santiago City, Philippines

TAGUINOD, VENTURITO B.

Address:
120 Panganiban St., City of Santiago, Isabela
Birthdate:
February 18, 1998
Phone:
0995-177-5230
Email:
venthrall@gmail.com

Skill Highlights

- Computer Literate
- Ability to work independently or as part of the group.
- Can work under pressure
- Willing to be trained for improvement

Language

English
Filipino

Objective Career

Technology Enthusiast; Well-versed in both computer and Information technology. Both a team-player and a leader; can accomplish tasks on time even under pressure, and eager to work with a company that shares the same values.

Trainings and Webinars

- "Web Development using NuxtJS AND Vue.js"
  College of Computing Studies, Information and Communication Technology (CCSICT) Isabela State University - Echague, Isabela - April 02, 2022
- "Flourishing the True Spirit of Genuine Leaders"
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  September 25-26, 2018
- ROADSHOW @ 9th: ROAD to Leading an Vibrant Country, Show it through the Engineering Technology
  Isabela State University, Echague Isabela
  March 15, 2018

Education

Secondary: University of La Sallette
High School Diploma
Santiago City, Philippines
May 2018

Tertiary: Cagayan Valley Computer and Information Technology College
Centro East, City of Santiago
June 2022

CVCITC

Cagayan Valley Computer and Information Technology College

#28 Carreon St., Centro East, Santiago City, Philippines

LAGUERTA, JESSA MAE

Address:
Malvar, City of Santiago, Isabela
Birthdate:
November 15, 1997
Phone:
0998-277-2443
Email:
jessamaelaguerta15@gmail.com

Skill Highlights

- Computer Literate: (MS Word, MS Excel, MS PowerPoint)
- Adobe Photoshop
- Great Communication Skill

Language

English
Filipino

Work Experiences

- SERVICE CREW at Mcdonald MartOne, Mabini Santiago City
- SERVICE CREW at Mang Inasal, Victory Norte Sabtiago City
- ASSISTANT SECRETARY at R&M Paint Center, Villasis Santiago City
- Graphic Artist at Ara G Captures Robinson's Mabini, Santiago City
- Graphic Artist at Best Picture Malvar, Santiago City
- Cashier at Red Ribbon Victory, Santiago City

Trainings and Webinars

- On-the-Job Training at San Andres, Santiago City
- Anti-Drugs Seminar at Sistech College Santiago City
- On-the-Job training at Cagayan Valley Computer and Information Technology College Inc.

Education

Primary: Mary Child Academy
Lawa, Meycauayan Bulacan
2004 - 2010

Secondary: Infant Jesus Montessori School
Malvar, Santiago City
2013-2014
Sistech College
Villasis Santiago City
2014-2016

Tertiary: Cagayan Valley Computer and Information Technology College
Centro East, City of Santiago
2019-2022

CVCITC

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#28 Carreon St., Centro East, Santiago City, Philippines

DE ASIS, RADLEY M.

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Purok 5, Villa Gonzaga City of Santiago, Isabela
Birthdate:
June 21, 1999
Phone:
0946-111-9295
Email:
radleydeasis06@gmail.com

Skill Highlights

- Computer Literate: (MS Word, MS Excel, MS PowerPoint)
- Electrical Servicing Skills
- Ability to work under pressure
- Good communication skills
- Good written and Verbal skill

Language

English
Filipino
Iloco

Work Experiences

- Engineering Office – Work Immersion at LGU Santiago City
  October – December 2016
- On-the-Job Training at PSCC LGU Santiago City
  February- April 2022

Trainings and Webinars

- "Web Development using NuxtJS AND Vue.js"
  College of Computing Studies, Information and Communication Technology (CCSICT) Isabela State University - Echague, Isabela - April 02, 2022
- "Flourishing the True Spirit of Genuine Leaders"
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  September 25-26, 2018
- ROADSHOW @ 9th: ROAD to Leading an Vibrant Country, Show it through the Engineering Technology
  Isabela State University, Echague Isabela
  March 15, 2018

Education

Primary: Villa Gonzaga Elementary School
Villa Gonzaga, Santiago City
2011 - 2012

Secondary: Patria Sable Corpus College
Calao East, Santiago City
2017-2018

Tertiary: Cagayan Valley Computer and Information Technology College
Centro East, City of Santiago
2019-2022

APPENDIX D. Source Code